AN INTELLIGENT SYSTEM FOR BIDIRCTIONAL SIGN LANGUAGE AND TEXT TRANSLATION

Synopsis submitted in the partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY
in
INFORMATION TECHNOLOGY

by

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CANDIDATE'S DECLARATION

We hereby declare that the work which is being presented here entitled, ***"An Intelligent system for bidirectional sign language and text translation*"** by **Tanishq, Narender, Chirag, Abhishek** in partial fulfillment of requirements for the award of the degree of B.Tech. in Information Technology submitted in the Department of Information Technology, PIET, Panipat affiliated to Kurukshetra University, is an authentic record of our own work under the supervision of **Ms. Rashmi Makkar**. The matter presented in this synopsis has not been submitted by us in full or in part to any other University / Institute for the award of B.Tech degree.

Signature of the Students Date: 28.01.2026

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This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

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Abstract

Communication barriers between hearing-impaired individuals and the hearing population pose a significant challenge in social, educational, and professional environments. Sign language is the primary medium of communication for deaf and mute individuals; however, its limited understanding among the general population restricts effective interaction. To address this issue, this project proposes **An Intelligent System for Bidirectional Sign Language and Text Translation**, aimed at enabling seamless two-way communication between sign language users and non-sign language users.

The system integrates computer vision, machine learning, and natural language processing techniques to perform real-time translation between sign language and text. For **Sign-to-Text** conversion, a camera-based approach is used to capture hand gestures, extract hand landmarks using MediaPipe, and recognize gestures through trained machine learning models, with a focus on Indian Sign Language (ISL). For **Text-to-Sign** conversion, the system translates textual or spoken input into corresponding sign language representations using images, GIFs, or videos. Additionally, speech-to-text functionality is incorporated to enhance usability.

The application is implemented as a web-based system using the Flask framework, providing an interactive and user-friendly interface. The proposed system is cost-effective, does not require specialized hardware, and supports real-time processing. It aims to promote inclusivity, accessibility, and digital empowerment by reducing communication barriers faced by the hearing-impaired community. The modular and scalable design allows future enhancements such as sentence-level gesture recognition, multilingual support, and mobile deployment.

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1. Introduction of the Problem and Objectives

Project Title Justification

The project titled **"An Intelligent System for Bidirectional Sign Language and Text Translation"** clearly represents the core objective of enabling two-way communication between sign language users and non-sign language users. The term bidirectional highlights the system's capability to perform both Sign-to-Text and Text-to-Sign translation, while intelligent system reflects the use of machine learning, computer vision, and natural language processing techniques. This title is precise, descriptive, and relevant to assistive technology and inclusive communication research.

General Introduction to the Problem

Communication plays a vital role in social interaction, education, and professional environments. However, hearing-impaired and speech-impaired individuals primarily rely on sign language, which is not widely understood by the general population. This communication gap often leads to social exclusion, reduced educational opportunities, and challenges in workplaces.

Existing solutions for sign language translation are limited, often focusing on only one direction of translation or lacking real-time accuracy. Many systems require specialized hardware such as sensor gloves, making them expensive and impractical for widespread use. Additionally, most available solutions do not support Indian Sign Language (ISL), which further limits accessibility for users in India.

This project addresses these challenges by developing a web-based intelligent system capable of translating sign language gestures into text and converting text or speech into sign language representations. By using camera-based gesture recognition and AI-driven models, the system provides a cost-effective, real-time, and user-friendly solution that enhances communication accessibility.

Specific Objectives

The project aims to achieve the following:

- Develop a real-time Sign-to-Text translation system using computer vision and machine learning.

- Implement Text-to-Sign translation using images, GIFs, or videos for clear sign representation.
- Support Indian Sign Language (ISL) gestures.
- Integrate speech-to-text functionality for enhanced usability.
- Build a web-based application using Flask for easy access and deployment.
- Enable multimedia input support such as recorded videos and YouTube links.
- Provide a scalable and modular system for future enhancements.

Justification of Objectives

This project deserves approval as it directly contributes to inclusive technology and social welfare by reducing communication barriers faced by the hearing-impaired community. Unlike traditional approaches that depend on costly hardware, this system uses a standard camera and software-based intelligence, making it affordable and accessible.

From an academic perspective, the project offers hands-on experience in machine learning, computer vision, natural language processing, and full-stack web development. It also aligns with current industry trends in assistive AI technologies. The project has strong real-world relevance and potential for future expansion into sentence-level translation, multilingual support, and mobile platforms.

Strengths and Limitation

Strengths:

- Supports bidirectional communication (Sign-to-Text and Text-to-Sign).
- Uses camera-based recognition without specialized hardware.
- Integrates AI, ML, and NLP technologies.
- Web-based and platform-independent.
- Modular architecture allows future enhancements.

Limitations:

- Recognition accuracy depends on lighting and camera quality.
- Limited vocabulary based on trained gestures.
- Continuous sentence-level sign recognition is complex.

2. Literature review

R.Rastgoo, K. Kiani, and S. Escalera [1] conducted a comprehensive survey on hand sign language recognition using deep learning. Their work reviews state-of-the-art techniques, **emphasizing that CNNs and RNNs have significantly improved recognition accuracy**, though challenges remain in handling variations across signers and environments.

Z. Zhang, J. Wu, and S. Zhao [2] proposed a **vision-based hand gesture recognition system** for sign language translation. The system employs computer vision techniques to capture hand movements and convert them into text or speech, achieving high accuracy in controlled environments, though limitations exist in handling complex sentence structures.

The International Journal of Computer Engineering [3] published research on Indian Sign Language recognition utilizing BERT and LSTM networks. This approach combines **natural language processing with temporal sequence modeling**, demonstrating that transformer-based architectures enhance recognition accuracy by capturing contextual relationships between signs.

Google's MediaPipe Hands [4] provides a real-time hand tracking framework that **detects 21 3D hand keypoints efficiently**. This open-source solution has become foundational in sign language recognition research due to its accuracy and ability to process video streams in real-time without specialized hardware.

A. Vaswani et al. [5] introduced the transformer architecture with attention mechanisms that have revolutionized **sequence learning**. While not designed for sign language, the transformer's ability to capture long-range dependencies makes it well-suited for understanding the sequential and spatial nature of sign language gestures.

A. Radford et al. [6] presented Whisper, **demonstrating effective large-scale pretraining and weak supervision** for speech recognition. Though focused on audio, the principles of transfer learning apply to sign language datasets, which often have limited labeled data.

The World Health Organization [7] documents that **millions worldwide rely on sign language for communication**. This establishes the social importance of sign language recognition technology as a critical tool for promoting inclusion and accessibility for the deaf and hard-of-hearing community.

Sr. No.	Author(s)	Research Focus	Technology Used	Outcomes	Limitations
[1]	R. Rastgoo, K. Kiani, S. Escalera	Deep learning approaches for hand sign language recognition	CNNs and RNNs for sign language	Comprehensive survey showing improved recognition accuracy with deep learning	Challenges in handling variations across different signers and environments
[2]	Z. Zhang, J. Wu, S. Zhao	Vision-based hand gesture recognition for translation	Computer vision techniques + feature extraction	High accuracy for limited vocabulary in controlled settings	Limited performance with complex sentence structures and contextual understanding
[3]	IJCESEN	Indian Sign Language recognition	BERT + LSTM networks	Enhanced recognition accuracy through contextual relationships between signs	Dataset limitations and complexity in real-world deployment
[4]	Google MediaPipe	Real-time hand tracking and gesture recognition	MediaPipe Hands framework (21 3D keypoints)	Efficient real-time processing without specialized hardware	Requires adequate lighting conditions and clear hand visibility
[5]	A. Vaswani et al.	Transformer architecture with attention mechanisms	Attention-based neural networks	Captures long-range dependencies in sequential data effectively	Not specifically designed for sign language, requires adaptation
[6]	A. Radford et al.	Large-scale weak supervision for speech recognition	Whisper model with transfer learning	Demonstrates effective pretraining strategies for limited data scenarios	Focus on audio rather than visual gesture recognition
[7]	WHO	Documentation of disability statistics and communication needs	Statistical analysis and global health data	Establishes social importance and scale of accessibility needs	Does not provide technical solutions, only contextual background

3. Methodology and Workflow

Requirement Analysis and Research (Week 1)

- Study existing sign language translation systems.
- Analyze Indian Sign Language gestures.
- Identify system requirements and limitations.

System Design and Environment Setup (Week 2)

- Configure Flask backend and project structure.
- Integrate MediaPipe for hand landmark detection.
- Load trained ML models for gesture recognition.

Sign-to-Text Module Development (Week 3–5)

- Capture hand gestures using camera input.
- Extract hand landmarks and preprocess data.
- Predict characters using trained ML models.

Text-to-Sign & Speech Integration (Week 6–7)

- Convert text input to sign images/videos.
- Implement speech-to-text using Whisper.
- Handle multimedia input (audio/video).

Testing and Optimization (Week 8–9)

- Validate gesture recognition accuracy.
- Optimize model performance and response time.
- Fix UI and integration issues.

User Interface and Deployment (Week 10)

- Design interactive web interface.
- Integrate APIs and real-time features.
- Prepare final documentation and demonstration.

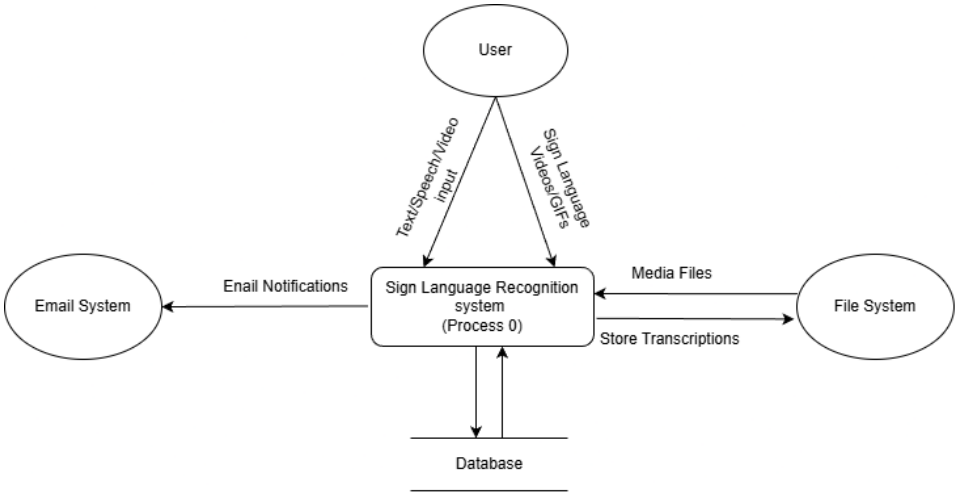


Fig 1. Level 0 Data flow Diagram

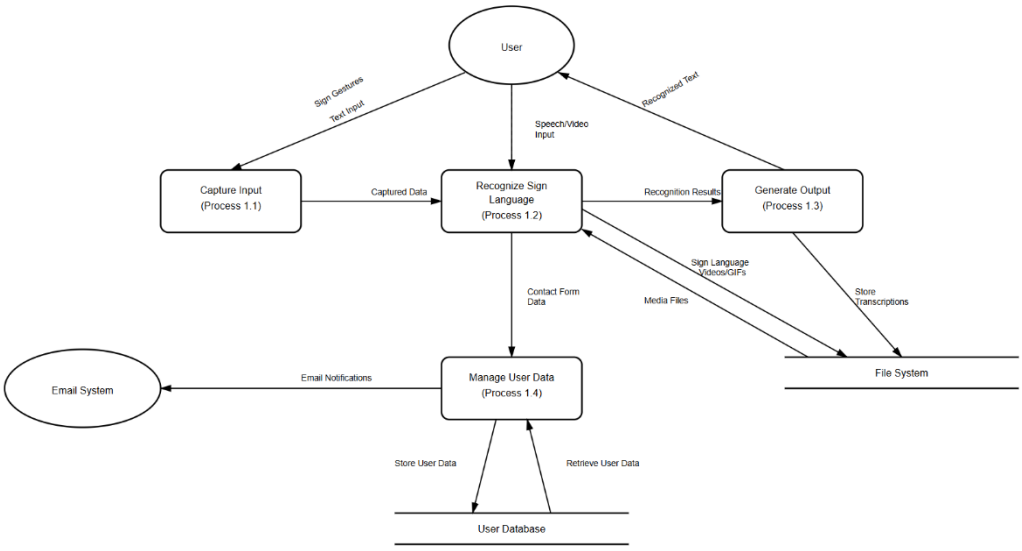


Fig 2. Level 1 Data Flow Diagram

4. Hardware and Software Requirements

Hardware Requirements:

- Processor: Intel Core i5 or higher
- RAM: 8GB or more (16GB recommended)
- Storage: 256GB SSD/HDD or more
- Graphics Card: Integrated or dedicated GPU
- Peripherals: Keyboard, Mouse, Monitor

Software Requirements:

- Operating System: Windows 10/11, Ubuntu 22.04 LTS, or macOS
- Development Environment: Visual Studio Code/ Sublime Text
- Programming Languages: HTML, CSS, JavaScript, Python
- Libraries/Frameworks: Flask, MediaPipe, TensorFlow, OpenCV, Whisper

5. Project Title Map with Engineering Program Outcomes and Program/Department Specific Outcomes

Project Title	**Engineering Program Outcomes (POs)**	**Department Specific Outcomes (DSOs)**
An Intelligent System for Bidirectional Sign Language and Text Translation	PO1, PO2, PO3, PSO5, PO9, PO10, PO12	DSO1, DSO2

6. Project Categorization

- **Application-based**: The project is aimed at developing an application to solve a specific problem or enhance a particular process.

Project Title	**Project Categorization**
An Intelligent System for Bidirectional Sign Language and Text Translation	Application based

7. Project Map with SDG Goals

The Project aligns with the following SDG goals:

- SDG 4: Quality Education – Democratizes web development learning by making website creation accessible to non-technical users while providing hands-on experience with frontend technologies

- SDG 8: **Decent Work and Economic Growth** – Enables small businesses and entrepreneurs to create professional websites without technical expertise, fostering digital entrepreneurship and economic opportunities.

- S**DG 9: Industry, Innovation, and Infrastructure** – Contributes to technological innovation through AI-driven automation in web development, advancing digital infrastructure capabilities.

8. Conclusions

The "**An Intelligent System for Bidirectional Sign Language and Text Translation"** project introduces an innovative assistive communication system that bridges the gap between hearing-impaired individuals and the general population. Unlike traditional sign language systems that focus only on one-way translation or require expensive hardware such as sensor gloves, this project emphasizes affordability, real-time processing, and software-based intelligence using computer vision and machine learning. By leveraging Flask, MediaPipe, deep learning models, and speech processing techniques, the system provides an interactive platform for converting sign language gestures into text and transforming text or speech into sign language representations.

The modular architecture of the system allows future enhancements such as sentence-level gesture recognition, multilingual sign language support, mobile application integration, and real-time conversational interfaces. The system also focuses on usability and accessibility, enabling users to learn, communicate, and interact effectively through sign language translation.

The positive impact of this project spans society, academics, and industry:

- **For Society:** The project promotes social inclusion by enabling effective communication between hearing-impaired individuals and non-sign language users, reducing communication barriers and enhancing accessibility in daily life.
- **For Academics:** It serves as an educational platform for students to understand machine learning, computer vision, artificial intelligence, and full-stack web development through a real-world assistive technology application.
- **For Colleges:** The system can be integrated into computer science and engineering curricula to provide practical exposure to AI-based systems, gesture recognition, and assistive technologies.

- **For Industry:** The project demonstrates the potential of AI-driven assistive communication tools and can be extended for use in healthcare, customer service, education platforms, and smart human-computer interaction systems.

9. Specific Contributions of Team Members

1. **Tanishq (Team Leader)**
    - **Role**: Project Management and AI Integration
    - **Role:** Backend Development and Model Training
    - **Leadership:** Oversees overall project planning, progress tracking, and ensures milestones and deadlines are achieved.
    - **Backend Development:** Designs and manages the Flask backend architecture, routing, API integration, and system logic.
    - **Model Training:** Responsible for training and integrating machine learning models for sign language recognition.
    - **AI Integration:** Implements computer vision and gesture recognition logic using hand landmark detection and classification models.
    - **Review:** Reviews, tests, and validates all modules to ensure smooth integration and reliable system performance.

2. **Narender (Backend Developer)**
    - **Role**: Backend Development and System Support
    - **Backend Logic**: Develops core Flask application structure and server-side functionalities.
    - **Backend Logic:** Assists in developing server-side functionalities using Flask.
    - **API Development:** Implements and manages RESTful APIs for Sign-to-Text and Text-to-Sign translation.
    - **Data Handling:** Manages data flow between frontend, backend, and trained ML models.
    - **System Integration:** Supports integration of speech-to-text, video processing, and multimedia features.

3. **Chirag (Frontend Developer and Research Analyst)**
    - **Role**: Frontend Design and Research Analysis
    - **Frontend Implementation:** Designs and develops the user interface for sign language translation modules.
    - **Research:** Conducts research on existing sign language translation systems and assistive technologies.
    - **User Experience:** Ensures intuitive interaction for gesture input, text input, and output display.
    - **Testing Support:** Assists in usability testing and interface optimization.

4. **Abhishek (Frontend Developer and Content)**

    - **Role**: Frontend Design and Content Development
    - **UI Design:** Develops visually appealing and accessible frontend layouts using HTML, CSS, and JavaScript.
    - **Backend Integration:** Integrates frontend components with backend APIs.
    - **User Experience:** Ensures smooth navigation and clarity of translated outputs (text, images, or videos).
    - **Documentation:** Assists in preparing project reports, documentation, and presentation materials.

10. Key References (In IEEE format only)

[1] R. Rastgoo, K. Kiani, and S. Escalera, "Hand Sign Language Recognition Using Deep Learning: A Survey," *Expert Systems with Applications*, vol. 164, pp. 113794, Feb. 2021, doi: https://doi.org/10.1016/j.eswa.2020.113794

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[6] A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision,” *Proceedings of the International Conference on Machine Learning (ICML)*, 2023. https://arxiv.org/abs/2212.04356

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